1]
Write a program to replace the each element of the array by sum of even digits present in that element. Take array element from user.
Example: a[5]={125,254,478,9854,254};
Output:---------={2,6,12,12,6};

2]
Write a program to reverse the alternate element of the array. Take array element from user.
Example:- a[5]={125,2547,7895,87,321}
Output: ---------{521,2547,5987,87,123}

3]
Write a program to add prime digit in each array element. Take array element from user.
Example: a[5]={2564,7845,1234,44,586}
Output:----------{7,12,5,0,5}

4]
Write a program to count armstrong element from the array. Take array element from user.
Example: a[5]={370,385,371,8,99}
Output:----count----3

5]
Write a program to reverse last(from right) three digit of each array element.Take array element from user. If the element is less than 3 digit then keep that element as it is.
Example:a[5]={3710,25,36987,15200,654}
Output---------{3017,25,36789 ,15002, 456}

6]
Write a program to reverse all the element in the array except largest element.
Take array element from user.
Example: a[5]={10,15,85,65,75};
Output:----------{1,51,85,56,57}

7]
Write a program to reverse first n element of the array. Take array element and n value from user.
Example:-----int a[5]={10,20,30,40,50} (if n=3 then reverse first three element only)
Output:--------int a[5]={30,20,10,40,50}

8]
Write a program to add all digits until first prime digit encounter from left. Take array element from user.
Example: a[5]={2443,12114,48755,8968,645}
Output:----------{11,6,10,31,0}

9]
Write a program to print binary of each array element using char * pointer.
Take array element from user.

10]
Write a program to print binary of each array element using short int * pointer.
Take array element from user.

11]
Write a program to store array element which contain even set bit count in another array. Take array element from user.
Example:----int a[5]={10,7,15,9,4}
Output:------int b[5]={10,15,9}

12]
Write a program to set nth ,(n+1)th, (n+2)th bit in each array element. Take array element and n value from user.
Example: a[5]={10,20,30,40,50} n=4 so set 4,5,6 bit in each array element.
Output:----------{122,116,126,120,122}

13]
Write a program to reverse the first prime element in an array from right. Take array element from user.
Example: a[5]={25,31,42,62,52} (31 is first prime element from right)
Output:--------- {25,13,42,62,52}

14]
Write a program to count the array element which are prime and at the same time there all digits also prime. Take array element from user.
Example: a[5]={254,333,443,577,523} (443,577,523 are prime but in 443 all digits are not prime)
Output-------Count=2

15]
Write a program to reverse the element at prime index. Take array element from user.
Example:---- a[5]={12,25,365,145,189} (365 is at index 2 which is prime)
Output:----------{12,25,563,541,189}

16]
Write a program to print binary of each array element using goto. Take array element from user.
Note: Use goto to print the binary.

17]
Write a program to reverse first three digits (from left)of each array element.
Take array element from user.
Example: int a[5]= {2546,254,54,8756,4569}
Output:---------- {4526,452,54,5786,6549}

18]
Write a program to delete prime digits from each array element. Take array element from user.
Example: int a[5]={2544,7895,3333,25876,2356}
Output---------- {44,89,0,86,6}

19]
Write a program to reverse the each array element after the first prime digit encounter from right in that array element. Take array element from user.
Example: int a[5]={425689,24689,8965,46789,54679}
Output:----------{245689,24689,6985,64789,64579}

20]
Write a program to replace each array element by the largest digit in that array element. Take array element from user.
Example: int a[5]={9856,2546,25863,2,2560}
Output:----------{9,6,8,2,6};

21]
Write a program to replace each array element by the set bit count in second byte of each array element. Count set bit count in second byte using short int pointer.
Take array element from user.
Example: int a[5]={512,12,768,256,360};
Output----------{1,0,3,1,1}

22]
Write a program to check first digit of each array element is prime or not.
If it is prime reverse that element.
Example: int a[5]={152,2569,856,428,512};
Output: a[5]={152,9652,856,428,215};

23]
Write a program to delete the prime number in a given array. Take array element from user.
Example: int a[6]={10,11,13,17,18,19}
Output----------{10, 18}

24]
Write a program to delete array element if it contain any one digit which contain even set bit count. Take array element from user.
Example: int a[6]={121,133,157,228,291,604} (here second element 133 contain 3 which contain even set bit count. Similarly 157 contain 5 which contain even set bit count so delete such element)
Output----------{121, 228, 604}

25]
Write a program to separate element with odd set bit count one side in the array (At starting) and element with even set bit count on the other side(At end of the array) in the same array. Take array element from user.
Example: int a[6]={15,13, 8, 5, 32 ,9} (Element in yellow contain even set bit count)
Output:----------{13,8,32,15 ,5 ,9}

26]
Write a program to separate prime element in one side of the array(At starting) and non prime element on the other side (At end) of the same array. Take array element from user.
Example:-------int a[6]={12,14,11,16,17 , 19}; (11,17,19 are pime element they should be at starting of the array)
Output:----------a[5]={11,17,19,12,14,16}

27]
Write a program to insert element in a array at a given position.
Take array element ,insert postion and inserting element from user.
Example:----- int a[6]={10,4,8,77,6} pos=3 num=100
Output:-------- a[6]={10,4,100,8,77,6}

28]
Write a program to insert element in a array at a postion next to the first prime number encounter in the array. The element which we have to insert that element should be fifth prime number from the first prime number encounter in the given array. Take array element from user.
Example:----------int a[6]={10 ,11 ,22,23,25} (In this array first prime number encounter is 11 so insert element next to 11 and the value of the element which we want to insert is 29 which is 5th prime number from 11)
Output:----------a[6]={10,11,29,22,23,25}

29]
Write a program to rotate array element toward left of the array. Take array element from user.
Example:---- int a[5]={10,20,30,40,50};
Output:----------a[5]={20,30,40,50,10};

30]
Write a program to rotate array element towards right of the array. Take array element from user.
Example:-------int a[5]={10,20,30,40,50};
Output:-------------a[5]={50,10,20,30,40};

31]
Write a program to rotate array element k times towards left of the array.
Take array element and k value from user.
Example: -----int a[5]={10,20,30,40,50}; k=3
Output:------------a[5]={40,50,10,20,30};

32]
Write a program to sort the array on the basis of data in the 3rd nibble.
If the data in the third nibble of any element is larger then that element should be at the last. Take array element from user.
Example: int a[5]={15<<8,15<<12,256,128,512};
Output: a[5]={15<<12,128,256,512,15<<8};

33]
Write a program to delete n bits from given position (pos) in each array element.
Take position (pos) and n from user.
Example: int a[5]={100,200,300,400,500}
Output:--------a[5]={12,24,36,48,60}

34]
Write a program to swap each nibbles in each byte of each array element. Take array element from user.

35]
Write a program to check data corresponding to first and third nibble is prime or not. If the data is prime corresponding to first and third nibble reverse that element.
Take array element from user.
Example: int a[5]={263,512,773,143,301} (binary of 263 is 0001 00000111 now data corresponding to first and third nibble is 00010111 which is 23 in decimal and 23 is prime then reverse that element)
Output:--------- a[5]={362,512,377,143,103}

36]
Write a program to sort the array on the basis of set bit count in each array element. That means element contain more set bit count it should be at last.
Take array element from user.
Example:---int a[5]={15,256,33,7,512}
Output:----------a[5]={256,512,33,7,15};
(256,512 have same number of set bits so if there sequence is reverse in output then also fine)

37]
Write a program to swap 2nd and 3rd nibble of each array element. Take array element from user.
Example:---int a[5]={0xABCD, 0x1234,0x5678,0x4567,0x7896)
Output:---------a[5]={0xACBD,0x1324,0x5768,0x4657,0x7986);

38]
Write a program to count the number of nibble in each array element whose value is prime. Check each nibble in each array element for prime value. Take array element from user.
Example:---- int a[5]={42,775,208,976,63}(42 binary is 00101010 here first nibble data is 10 which is not prime but second nibble data is 2 which is prime so prime nibble count is 1 so output for first element is 1)
Output:----------42----1
775----2
208----1
976----2
63----1

39]
Write a program to extract the bits from position 3 to 10 of each array element and check whether the extracted value is prime or not . If the extracted value is prime then replace the whole array element with extracted value.
Take array element from user.
Example:---- int a[5]={100,200,300,350,400} (binary of 100 is 00000***000 01100***100 extract data from postion 3 to 10 is 00001100 and this data is not prime. Similarly binary of 200 is 0000 11001000 extracted data from position 3 to 10 is 00011001 which is 25 in decimal and it is also not prime so 200 will remain as it is)
Output:------int a[5]={100, 200, 37,43,400}

40]
Write a program to swap first and second byte of each array element. Take array element from user.
Example:-----int a[5]={0xABCD, 0x1234, 0xDEF1,0xAB12}
Output:--------int a[5]={0xCDAB,0x3412,0xF1DE,0x12AB}

41]
Write a program to check data in third nibble is prime or not. If the data in third nibble is prime clear the third nibble. Take array element from user.
Example:--------int a[5]={777,525,300,800,100}
Output:----------int a[5]={ 9, 13,300,32, 100}

42]
Write a program to delete first digit of each array element if that digit is prime. Take array element from user.
Example: Int a[5]={123,245,312,599,600};
Output : a[5]={123,45,12,99,600};

43]
Write a program to replace each array element by the nth prime number from the given array element. Take array element from user.
Here n is number of prime digit in a given number.
Example: int a[5]={12,2,3,4,5};(in 12 prime digit count is 1 so 1st prime number from 12 is 13)
Output: a[5]={13,3,5,4,7};

44]
Write a program to print binary of each element of a double data type array using short int pointer. Take array element from user.

45]
Write a program to swap first digit and last digit in each array element if both digit contain even set bit count. Take array element from user.
Example: int a[5]={9245,3476, 1254,3456,4545}
Output:-----int a[5]={5249,6473 ,1254, 6453,4545}

46]
Write a program to accept an integer array from the user. For each element of the array find the 5th prime number from that array element and replace the element with that 5th prime number.
Take array element from user.
Example:------int a[5]={5,10,15,20,25};
Output:---------int a[5]={19,23,31,41,43};

47]
Write a program to reverse only prime digit in each array element while keeping non prime digit at original location. Take array element from user.
Example:----int a[5]={12687, 24513, 45678,45738,2307};
Output :----int a[5]={17682 ,34512 ,47658,43758,7302};

48]
Write a program to reverse first two prime number from last in an given array.
Take array element from user.
Example:------int a[5]= {19,15,13,14,17}
Output:-------------a[5]={19,15,31,14,71}

49]
Write a program to add largest digit of each array element. Take array element from user.
Example:-----int a[5]={258,456,725,89645,33} (largest digit in 258 is 8, similarly largest digit in 456 is 6 so add all these largest digits)
Output:-------8+6+7+9+3-----33

50]
Write a program to swap first prime number from left and first prime number from right in a given array.
Example:----int a[5]={12,11,19,17,25}; (From right first prime number is 17 and from left first prime number is 11. So swap them both)
Output:--------int a[5]={12,17,19,11,25};

51]
Write a program to check first four bits of each array element.
If all the first four bits are 1 then clear all four bits in that element.
If all the first four bits are 0 then set all all the four bits in that element.
If some bit are zero and some bits are 1 then toggle all four bits.
Example:-----int a[5]={266,271,256,263,240} (In 266 binary 0001 00001010 Out of first four bit some bits are zeo some bits are 1 so toggle all the bits. Similarly 271 binary is 0001 00001111 all first four bits are 1 so clear all bits. In 256 all first four bits are 0 then set all the bits.)
Output:-----int a[5]={261,256,271,264,255}

52]
Write a program to replace each element of the array by count and that count should represent how many prime number are there before that cooresponding element.
For example
Input int a[5]={10,20,30,40,50};
Output a[5]={4,8,10,12,15}
where 4 is number of prime number before 10 and so on.

53]
Write a program to print following pattern using array.
Example: int a[5]={1,2,3,4,5};

```
        1
      2 1
    3 2 1
  4 3 2 1
5 4 3 2 1
  4 3 2 1
    3 2 1
      2 1
        1
```

Note: Use array to print the content of the pattern.